Software Requirements Specification: Football Analytics Dashboard

This SRS defines what a Python and Streamlit dashboard must do, which data and software it needs, and how each requirement will be verified. Version 1.0 (draft).

1. Introduction

1.1 Purpose

This document specifies the software requirements for the Football Analytics Dashboard, a web app that turns raw Premier League data into interactive charts. Its readers are the developers who build it, the testers who verify it and the reviewers who approve it.

1.2 Scope

The system loads a cleaned football dataset, lets users filter it by season, team and player, and shows team and player performance as interactive charts. It also compares two players and generates short text insights. It is deployed publicly and opened with a link.

Out of scope for version 1:

- Live or real-time data feeds.
- Event or tracking data (pass networks, shot maps, player positions).
- User accounts, saved preferences or any stored user data.
- Native mobile apps.

1.3 Definitions

Term	Meaning
SRS	Software Requirements Specification, this document
FR / NFR / DR	Functional, non-functional and data requirement IDs
Streamlit	Python framework that turns a script into a web app

Term	Meaning
Plotly	Python library for interactive charts
Radar chart	Chart that plots several metrics of one player on spokes from a common centre
Per-90	A statistic divided by minutes played and multiplied by 90, so players with different playing time can be compared
Cascading filter	A filter whose options depend on the choice made in another filter
MoSCoW	Priority scale: Must, Should, Could

2. Overall Description

2.1 Product perspective

The dashboard is a standalone, single-page web app. At runtime it reads cleaned CSV files from its own repository. It uses no database and calls no external API, which keeps hosting free and behavior predictable.

2.2 User classes

User	Technical skill	Main need
Casual fan	None	See how a team or player performed in a few clicks
Student or analyst	Basic	Compare players and teams, spot trends
Developer or maintainer	Python	Add a season, fix data issues, extend charts

2.3 Operating environment

- **Client:** the latest two versions of Chrome, Firefox, Safari and Edge, on desktop and mobile.

- **Server:** Python 3.10 or newer, hosted on Streamlit Community Cloud.
- **Source:** a public GitHub repository linked to the host.

2.4 Constraints

- The free hosting tier has limited memory and CPU, so the app must stay lightweight and cache its data.
- The dataset license must allow public redistribution, with attribution to the source.
- Version 1 covers one league (Premier League) and one full season.

2.5 Assumptions and dependencies

- A dataset with player-level statistics (goals, assists, minutes and similar) is available. Match-level data alone cannot support player features.
- Match results come from a match-level file (for example football-data.co.uk) if team analysis is built on matches rather than player totals.
- **Open item:** the final dataset is not yet selected. Field names in section 4 are the target schema and will be mapped to the real columns.

3. Functional Requirements

Priority uses MoSCoW: Must is required for release, Should is expected, Could is built only if time allows.

3.1 Data loading and cleaning

ID	Requirement	Priority
FR-01	The system shall load the cleaned CSV files from `data/clean/` at startup and cache them so filter changes do not re-read the disk.	Must
FR-02	The system shall show a readable error message, not a stack trace, if a data file is missing or lacks a required column.	Must
FR-03	The system shall map team names through a lookup table so each club appears under exactly one name.	Must

ID	Requirement	Priority
FR-04	The system shall derive calculated fields (result per team, points, goal difference, per-90 statistics) during cleaning, not at display time.	Should

3.2 Filters

ID	Requirement	Priority
FR-05	The team filter shall build its options from the loaded data, with no hardcoded team names.	Must
FR-06	The season filter shall build its options from the data and be hidden when only one season is loaded.	Should
FR-07	The player filter shall list only players of the selected team and season (cascading).	Must
FR-08	A venue filter (All, Home, Away) shall apply to team analysis.	Should
FR-09	Every chart, metric and insight on screen shall update on the same rerun when any filter changes, with no separate apply button.	Must
FR-10	The system shall show a "No data for this selection" message when a filter combination returns zero rows.	Must

3.3 Season overview

ID	Requirement	Priority
FR-11	The system shall display matches played, number of teams, total goals and average goals per match for the current selection.	Must

ID	Requirement	Priority
FR-12	The system shall display the most successful team, defined as most points, with ties broken by goal difference and then goals scored.	Must

3.4 Top performers

ID	Requirement	Priority
FR-13	The system shall show the top 10 goal scorers as a horizontal bar chart with player, team and goals.	Must
FR-14	The system shall show the top 10 assist providers in the same format.	Must
FR-15	The system shall show the full ranking as a sortable table below each chart.	Should

3.5 Team analysis

ID	Requirement	Priority
FR-16	The system shall show the selected team′s wins, draws and losses as a stacked bar, split by home and away when the venue filter is All.	Must
FR-17	The system shall show cumulative points across matchdays as a line chart.	Must
FR-18	The system shall show goals scored and conceded per match.	Should
FR-19	Chart tooltips shall show the opponent, score and date of each match.	Should

3.6 Player comparison

ID	Requirement	Priority
FR-20	The user shall select two players from dropdowns, and the system shall prevent selecting the same player twice.	Must
FR-21	The system shall draw a radar chart of at least five metrics, each normalized to a 0 to 100 scale (percentile among all players above a minimum-minutes threshold).	Must
FR-22	The system shall draw a grouped bar chart of the raw values of the same metrics.	Must
FR-23	The system shall show each player's team, position, matches and minutes beside the charts.	Should
FR-24	The system shall warn the user when a selected player has fewer minutes than the threshold (for example 450) or a metric is unavailable.	Could

3.7 Insights

ID	Requirement	Priority
FR-25	The system shall generate at least five plain-language insights from the data, such as top scorer, top assist provider, best home record, highest-scoring match and average goals per match.	Must
FR-26	Insights shall recalculate when filters change.	Should

3.8 Match winner prediction (optional)

ID	Requirement	Priority
FR-27	The system shall train a classifier on past matches and show the predicted outcome (home win, draw, away win) with probabilities for a chosen fixture.	Could

ID	Requirement	Priority
FR-28	The system shall report the model's accuracy on a held-out test set split by date, never by random shuffle.	Could

4. Data Requirements

The dataset shall cover one full Premier League season (380 matches, 20 teams) with player-level statistics. The tables below give the target schema; real column names are mapped during cleaning.

4.1 Match data

One row per match.

Field	Type	Required	Notes
Date	date	Yes	Parsed to datetime
HomeTeam, AwayTeam	text	Yes	Standardized names
FTHG, FTAG	integer	Yes	Full-time home and away goals
FTR	text (H, D, A)	Yes	Derived from goals if missing
HS, AS, HST, AST	integer	No	Shots and shots on target
HC, AC	integer	No	Corners
HY, AY, HR, AR	integer	No	Yellow and red cards

4.2 Player data

One row per player per season.

Field	Type	Required	Notes
Player	text	Yes	Unique per player and season
Team	text	Yes	Standardized to match section 4.1

Field	Type	Required	Notes
Position	text	Yes	Grouped to GK, DF, MF, FW
Matches, Minutes	integer	Yes	Used for per-90 values and the minimum-minutes rule
Goals, Assists	integer	Yes	Primary metrics
Shots, ShotsOnTarget	integer	No	Radar metrics
Passes, PassAccuracy	integer, percent	No	Radar metrics if available
YellowCards, RedCards	integer	No	Discipline

4.3 Data quality rules

ID	Rule
DR-01	Each club shall appear under one name across all files.
DR-02	Rows with a missing date, team, player name or goals value shall be dropped and counted in a cleaning log.
DR-03	Duplicate rows shall be removed.
DR-04	Numeric columns shall be stored as numbers, not text.
DR-05	Optional missing values shall be marked as unavailable, not silently set to zero, except where zero is the true value.
DR-06	Total goals from the match file shall equal the sum of the home and away goals in the raw source.
DR-07	Cleaned files shall be saved in `data/clean/` and the raw files kept unchanged in `data/raw/`.

5. Non-Functional Requirements

ID	Category	Requirement	How it is measured
NFR-01	Performance	First page load shall complete within 5 seconds on the hosted app once it is running, and any filter change shall redraw within 2 seconds.	Manual timing, 10 runs
NFR-02	Scalability	The app shall support up to five seasons (about 1,900 matches) without filter updates exceeding 3 seconds.	Load test with five seasons
NFR-03	Usability	A first-time user shall complete the tasks "find the top scorer" and "compare two players" without instructions. Every chart shall have a title, labeled axes and units.	Test with three users
NFR-04	Responsiveness	The layout shall be readable at 1280 px (desktop) and 390 px (phone) widths.	Visual check at both widths
NFR-05	Accessibility	Text contrast shall meet 4.5:1, and charts shall not rely on color alone to separate series.	Contrast checker, grayscale view
NFR-06	Compatibility	The app shall work in the latest two versions of Chrome, Firefox, Safari and Edge.	Manual test per browser
NFR-07	Reliability	No combination of filters shall crash the app. Errors shall appear as messages.	Test of all filter combinations
NFR-08	Maintainability	Data loading, cleaning, charts and insights shall live in separate modules. Functions shall have docstrings. Settings (file paths, thresholds) shall sit in one place.	Code review

ID	Category	Requirement	How it is measured
NFR-09	Portability	A clean install with `pip install -r requirements.txt` and one run command shall start the app.	Fresh virtual environment test
NFR-10	Privacy and licensing	The app shall collect no personal user data and keep no secrets in the repository. The dataset source and license shall be credited in the app footer and README.	Repository review
NFR-11	Data accuracy	Figures shown in the app shall match values computed directly from the cleaned data.	Spot checks, see section 9

6. External Interface Requirements

6.1 User interface

ID	Requirement
UI-01	A sidebar shall hold the season, team, player and venue filters.
UI-02	The main area shall use tabs: Overview, Top Performers, Team Analysis, Player Comparison, Insights.
UI-03	Each team or player shall keep the same color across all charts.
UI-04	Charts shall support hover tooltips, zoom reset and PNG export (default Plotly toolbar).
UI-05	Wide layout shall be used, with metric cards above the charts.

6.2 Software interfaces

- **Pandas to files:** CSV read with `pandas.read_csv`, cached with `st.cache_data`.

- **Plotly to Streamlit:** figures are rendered with `st.plotly_chart`.
- **GitHub to host:** the hosting service pulls `app.py` and `requirements.txt` from the repository on each push.

6.3 Hardware interfaces

None. Users need a device with a modern browser. Developers need any machine that runs Python 3.10 or newer.

7. Software and Environment Requirements

Component	Requirement	Purpose
Python	3.10 or newer	Runtime
pandas	2.0 or newer	Loading, cleaning, aggregation
numpy	1.24 or newer	Numeric operations, normalization
Streamlit	1.30 or newer	Web app, widgets, caching
Plotly	5.18 or newer	Interactive bar, line and radar charts
scikit-learn	1.3 or newer (only for FR-27 and FR-28)	Prediction model
pytest	Any current version	Automated tests
Git and GitHub	Any current version	Version control and deployment source

Exact versions shall be pinned in `requirements.txt` when the project is built, so the hosted app and local runs behave identically. Example:

```
pandas==2.2.2
numpy==1.26.4
streamlit==1.35.0
plotly==5.22.0
scikit-learn==1.5.0
```

The versions above are examples to be confirmed at build time.

8. System Architecture

The app has four layers. Each layer only calls the one below it.

1. **Presentation** (`app.py`): page layout, sidebar widgets, tabs.
2. **Chart and insight logic** (`charts.py` , `insights.py`): turn a filtered table into a Plotly figure or a sentence.
3. **Data access** (`load_data.py`): read, cache and filter the cleaned tables.
4. **Data preparation** (`clean_data.py`): run once, offline, from raw to clean files.

Project structure:

```
football-dashboard/
  app.py
  config.py              # paths, thresholds, color map
  data/
    raw/
    clean/
  src/
    load_data.py
    clean_data.py
    charts.py
    insights.py
    model.py             # optional
  tests/
    test_clean_data.py
    test_insights.py
  requirements.txt
  README.md
```

9. Deployment Requirements

ID	Requirement
DEP-01	The code and cleaned data shall be in a public GitHub repository.
DEP-02	The repository shall contain `requirements.txt` and a README with setup, run and data-source instructions.

ID	Requirement
DEP-03	The app shall deploy to Streamlit Community Cloud with `app.py` as the entry point and no manual server setup.
DEP-04	The deployed link shall open without login.
DEP-05	Pushing to the main branch shall update the live app.

Local run command: `streamlit run app.py`.

10. Verification and Acceptance

A requirement is accepted when its test passes. Tests marked automated run with pytest; the rest are manual checks.

Test	Covers	Pass condition	Type
T-01	FR-01, FR-02, DR-02	App starts with valid data; missing file or column shows a message	Automated
T-02	FR-03, DR-01	Unique team count equals 20, with no near-duplicate names	Automated
T-03	FR-05, FR-07	Team list equals the teams in the data; player list shows only the selected team	Automated
T-04	FR-09, FR-10	Changing any filter updates all charts; an empty selection shows the no-data message	Manual
T-05	FR-11, FR-12, NFR-11, DR-06	Overview totals and the top team match values computed independently from the raw file	Automated
T-06	FR-13, FR-14	Top 10 lists match a sorted check of the cleaned data	Automated
T-07	FR-16, FR-17	Wins + draws + losses equals matches played for every team; final cumulative points match the league table	Automated

Test	Covers	Pass condition	Type
T-08	FR-20 to FR-22	The same player cannot be picked twice; radar values stay between 0 and 100	Automated
T-09	FR-25, FR-26	At least five insights appear and change with the filters	Manual
T-10	NFR-01, NFR-02	Timing targets met	Manual
T-11	NFR-04, NFR-06	Layout readable at 1280 px and 390 px in four browsers	Manual
T-12	NFR-09, DEP-01 to DEP-05	Fresh environment installs and runs; public link opens	Manual
T-13	FR-27, FR-28	Test split is by date; accuracy is reported	Automated

11. Open Issues

ID	Issue	Needed to close
OI-01	Final dataset not selected	Dataset name or its column list
OI-02	Minimum-minutes threshold for the radar percentile is only an example (450)	Decision after viewing player data
OI-03	Whether team analysis uses match-level data or aggregates from player data	Confirmed by the chosen dataset
OI-04	Whether the optional prediction model (FR-27, FR-28) is built	Time left after core features